KOPS Multi-Node Kubernetes Cluster Setup on AWS (2026 Updated)

What is KOPS?

KOPS (Kubernetes Operations)

- **Production-grade Kubernetes cluster setup tool**
- **Mainly used on AWS**
- **Automates:**
  - **EC2**
  - **VPC**
  - **ASG**
  - **Load Balancer**
  - **Worker Nodes**
  - **Master Nodes**

Advantages

- **Automates infra creation**
- **HA Kubernetes setup**
- **Rolling updates**
- **Multi-node cluster support**
- **Production-ready clusters**

Alternatives

- **EKS**
- **Minikube**
- **kubeadm**
- **Rancher**
- **Terraform**

Architecture

Component	Count
Master Node	1
Worker Nodes	2
Region	ap-south-1
Instance Type	t3.medium / t3.small

STEP 1 — Launch EC2 Instance

Launch:

Amazon Linux 2023

or

Ubuntu 24.04 LTS

Recommended:

Resource	Value
Type	t3.medium
Storage	20 GB
IAM Role	AdministratorAccess
Security Group	Allow All Traffic (Lab Only)

STEP 2 — Connect to EC2

```
ssh -i key.pem ec2-user@<PUBLIC-IP>
```

Ubuntu:

```
ssh -i key.pem ubuntu@<PUBLIC-IP>
```

STEP 3 — Install AWS CLI (2026)

Amazon Linux

sudo dnf install unzip curl wget git tar -y

curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"

unzip awscliv2.zip

sudo ./aws/install

Verify

aws --version

STEP 4 — Install kubectl (2026)

curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"

chmod +x kubectl

sudo mv kubectl /usr/local/bin/

Verify

kubectl version --client

STEP 5 — Install KOPS (2026)

Latest Stable KOPS

curl -LO https://github.com/kubernetes/kops/releases/latest/download/kops-linux-amd64

chmod +x kops-linux-amd64

sudo mv kops-linux-amd64 /usr/local/bin/kops

Verify

kops version

STEP 6 — Configure PATH (Optional)

vi ~/.bashrc

Add:

export PATH=$PATH:/usr/local/bin

Apply:

source ~/.bashrc

> Ubuntu usually does not require this step.

STEP 7 — Create S3 Bucket for Cluster State Store

Important

KOPS stores cluster configuration in S3.

Create Bucket

**aws s3api create-bucket **

**--bucket reyaz-kops-testbkt123.k8s.local **

**--region ap-south-1 **

--create-bucket-configuration LocationConstraint=ap-south-1

Enable Versioning

**aws s3api put-bucket-versioning **

**--bucket reyaz-kops-testbkt123.k8s.local **

**--region ap-south-1 **

--versioning-configuration Status=Enabled

Export KOPS State Store

export KOPS_STATE_STORE=s3://reyaz-kops-testbkt123.k8s.local

Permanent Export

echo 'export KOPS_STATE_STORE=s3://reyaz-kops-testbkt123.k8s.local' >> ~/.bashrc

source ~/.bashrc

STEP 8 — Create Kubernetes Cluster

Updated 2026 Recommended Command

**kops create cluster **

**--name=reyaz.k8s.local **

**--cloud=aws **

**--zones=ap-south-1a **

**--control-plane-count=1 **

**--node-count=2 **

**--control-plane-size=t3.medium **

**--node-size=t3.small **

**--networking=calico **

**--topology=public **

--ssh-public-key ~/.ssh/id_rsa.pub \

--yes

Important Fixes From Images

Old	Updated 2026
`--master-count`	`--control-plane-count`
`--master-size`	`--control-plane-size`
`t2.micro`	`t3.small / t3.medium`
old KOPS download	latest stable download
old AMI assumptions	Amazon Linux 2023 / Ubuntu 24

STEP 9 — Validate Cluster

kops validate cluster --wait 10m

STEP 10 — Verify Nodes

kubectl get nodes

Expected:

NAME	STATUS	ROLES	AGE
i-xxxxx	Ready	control-plane	
i-yyyyy	Ready	node	
i-zzzzz	Ready	node	

STEP 11 — Verify Pods

kubectl get pods -A

STEP 12 — List Clusters

kops get cluster

STEP 13 — Edit Cluster

kops edit cluster reyaz.k8s.local

STEP 14 — Edit Node Group

Worker Nodes

kops edit ig --name=reyaz.k8s.local nodes-ap-south-1a

Control Plane

kops edit ig --name=reyaz.k8s.local control-plane-ap-south-1a

---

STEP 15 — Apply Changes

kops update cluster --name reyaz.k8s.local --yes --admin

STEP 16 — Validate Again

```
kops validate cluster --wait 10m
```

STEP 17 — Kubernetes Test

```
kubectl create deployment nginx --image=nginx
```

```
kubectl get pods
```

STEP 18 — Delete Cluster (IMPORTANT)

To avoid AWS billing:

```
kops delete cluster --name reyaz.k8s.local --yes
```

Useful Commands

Cluster Info

```
kubectl cluster-info
```

Get All Resources

kubectl get all -A

Watch Nodes

kubectl get nodes -w

Security Group Ports

Port	Purpose
22	SSH
6443	Kubernetes API
443	HTTPS
80	HTTP

Beginner Notes

KOPS Creates Automatically

- **VPC**
- **EC2**
- **Auto Scaling Groups**
- **Load Balancers**
- **IAM resources**
- **Security Groups**

Important 2026 Notes

- **Use `t3` instances instead of `t2`**
- **Use latest KOPS release**
- **Use Calico networking**
- **Use Amazon Linux 2023 or Ubuntu 24**
- **Always delete cluster after lab/testing**

KOPS (Kubernetes Operations)

What is KOPS?

Faculty Definition:

Kops (Kubernetes Operations) is a command-line tool that helps you create, manage, upgrade, and destroy Kubernetes clusters on cloud providers like AWS, GCP and more.

Simple Meaning:

- KOPS is a tool used to create Kubernetes clusters automatically.
- Mostly used on AWS.
- It creates production-level Kubernetes clusters.

Easy Real-Time Understanding

Without KOPS:

You manually create:
- EC2 instances
- Networking
- Master nodes
- Worker nodes
- Security groups
- Kubernetes setup

This is:

Very difficult and time consuming.

With KOPS:

Single command can create complete Kubernetes cluster.

KOPS automates:

- Infrastructure setup
- Cluster creation
- Node creation
- Networking
- HA setup

Important Point

Faculty Statement:

KOPS will just create Production level Cluster, it's a 3rd party tool.

Simple Meaning:

- KOPS is not directly part of Kubernetes.
- It is an external tool.
- Used for real-world production environments.

Why KOPS is Popular?

Faculty Definition:

It's often referred to as the "kubectl" for clusters.

Simple Meaning:

- kubectl manages Kubernetes resources.
- KOPS manages Kubernetes clusters.

Advantages of KOPS

1. Automates Infrastructure Creation

Faculty Definition:

Automate infra creation in AWS and GCE Kubernetes cluster.

Simple Meaning:

- Automatically creates AWS resources.

Example:

EC2
VPC
Subnets
IAM
Security Groups

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2. Deploys HA Kubernetes Master

Faculty Definition:

Deploys HA K8S master

Simple Meaning:

- Creates Highly Available masters.

Why?

If one master fails, another continues working.

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3. Supports Rolling Updates

Faculty Definition:

Supports rolling updates

Simple Meaning:

- Update cluster without downtime.

Example:

Upgrade Kubernetes version safely.

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4. Supports Homogeneous and Heterogeneous Clusters

Faculty Definition:

supports homogenous and heterogenous clusters

Simple Meaning:

Homogeneous Cluster

Same node types.

Example:

All nodes are t2.micro

Heterogeneous Cluster

Different node types.

Example:

Master = t2.medium
Workers = t2.micro

---

5. Saves Time and Work

Faculty Definition:

save time and work

Simple Meaning:

- Reduces manual setup effort.
- Faster cluster creation.

Alternatives to KOPS

Faculty Notes:

EKS
MINIKUBE
KUBEADM
RANCHER
TERRAFORM

Explanation of Alternatives

Tool	Purpose
Minikube	Local single-node Kubernetes
kubeadm	Manual Kubernetes setup
EKS	Managed Kubernetes by AWS
Rancher	Kubernetes management platform
Terraform	Infrastructure automation

Important Difference

KOPS vs kubeadm

KOPS	kubeadm
Automates cluster creation	Manual setup
Mostly cloud-based	Manual configuration
Production-ready	Learning + manual production
Easier	More complex

KOPS Setup

Step 1: Launch EC2 Instance

Faculty Note:

Launch Amazon Linux 2 EC2 instance with t2.micro

Simple Meaning:

- Create AWS EC2 server.

Recommended:

Amazon Linux 2
t2.micro

Step 2: Attach IAM Role

Faculty Note:

Attach IAM Role with Admin permissions

Why?

KOPS needs permission to create AWS resources.

Step 3: Install AWS CLI

Important:

AWS CLI must be installed.

Purpose:

- Allows terminal to communicate with AWS.

Step 4: Install kubectl and kops

Faculty Commands:

curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"

wget https://github.com/kubernetes/kops/releases/download/v1.25.0/kops-linux-amd64

chmod +x kops-linux-amd64 kubectl

mv kubectl /usr/local/bin/kubectl

mv kops-linux-amd64 /usr/local/bin/kops

Command Explanation

Download kubectl

curl -LO ...

Purpose:

- Downloads kubectl binary.

Download KOPS

wget ...

Purpose:

- Downloads KOPS binary.

Give Execute Permission

chmod +x

Purpose:

- Makes files executable.

Move to System Path

mv kubectl /usr/local/bin/kubectl

Purpose:

- Allows kubectl command from anywhere.

Step 5: Update PATH Variable

Faculty Notes:

vi .bashrc

Add:

export PATH=$PATH:/usr/local/bin/

Save:

wq!

Apply Changes:

source .bashrc

Why PATH is Needed?

Faculty Definition:

Appends /usr/local/bin/ to system PATH variable,
allowing you to execute binaries from anywhere.

Simple Meaning:

- Linux searches commands using PATH.
- Adding `/usr/local/bin` allows:

kubectl
kops

to run globally.

Important Faculty Note

In Ubuntu no need to do above steps,
but for Amazon Linux yes.

Simple Meaning:

- Ubuntu already contains correct PATH.

- Amazon Linux may require manual configuration.

Real-Time Architecture Understanding

KOPS
↓
Creates Kubernetes Cluster
↓
Master Nodes
Worker Nodes
Networking
Storage
IAM

KOPS Main Use Case

Faculty Note:

How to work on Multi-Node cluster servers

Simple Meaning:

- KOPS is mainly used to create:

Multi-node Kubernetes clusters.

Single Node vs Multi Node

Single Node	Multi Node
One machine	Multiple machines
Learning purpose	Production purpose
Minikube	KOPS

Important Interview Question

Question:

What is KOPS?

Answer:

KOPS is a Kubernetes Operations tool used to create, manage and automate production-grade Kubernetes clusters, mainly on cloud providers like AWS.

Important Interview Question

Question:

Why do we use KOPS?

Answer:

KOPS automates Kubernetes cluster creation and infrastructure setup.
It reduces manual configuration effort.

Important Interview Question

Question:

What is the difference between KOPS and kubectl?

Answer:

kubectl	KOPS
Manages Kubernetes resources	Manages Kubernetes clusters
Works inside cluster	Creates entire cluster

Important Interview Question

Question:

Why attach IAM Role to EC2?

Answer:

KOPS requires AWS permissions to create infrastructure resources.

Important Interview Question

Question:

Why use chmod +x?

Answer:

It gives executable permission to binary files.

Important Interview Question

Question:

Why move binaries to /usr/local/bin?

Answer:

To access commands globally from terminal.

Final Quick Revision

KOPS:
- Kubernetes Operations tool
- Used to create production Kubernetes clusters
- Mostly used on AWS

- Automates infrastructure creation

Advantages:
- HA masters
- Rolling updates
- Saves time
- Production-ready

Important Commands:
- curl
- wget
- chmod +x
- mv
- source .bashrc

Alternatives:
- EKS
- kubeadm
- Minikube
- Rancher
- Terraform